

GANYUSHKINO, Kazakhstan

WMO: 347980

Lat: 46.6053N

Lon: 49.2381E

Elev: -23

StdP: 101.60

Time Zone: 5.00 (E05)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.3	-17.9	-24.2	0.4	-20.8	-21.0	0.6	-17.5	14.3	-4.8	13.8	-3.7	2.7	90	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	36.1	20.9	34.6	20.6	33.2	20.3	23.4	31.1	22.4	30.6	21.5	30.0	4.7	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.0	15.6	27.8	19.6	14.3	26.5	18.5	13.3	26.1	69.6	31.0	65.6	30.7	62.5	30.0	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.5	11.1	9.3	DB	-23.8	38.4	4.4	1.6	-27.0	39.6	-29.6	40.5	-32.0	41.4	-35.3	42.6	(4)
(5)			WB	-23.8	25.2	4.3	1.6	-26.9	26.3	-29.5	27.2	-31.9	28.1	-35.0	29.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	-5.5	-4.8	2.8	11.0	19.1	23.9	26.1	24.7	18.1	10.3	2.8	-2.4	(6)	
	DBStd	12.15	6.13	5.96	4.18	4.19	3.96	3.35	2.77	3.49	3.93	4.37	5.03	5.27	(7)	
	HDD10.0	1811	480	416	225	37	1	0	0	0	1	51	217	384	(8)	
	HDD18.3	3541	738	649	482	221	38	3	0	2	50	250	465	643	(9)	
	CDD10.0	2030	0	0	2	67	283	418	498	457	243	59	3	0	(10)	
	CDD18.3	718	0	0	0	1	62	171	240	201	43	1	0	0	(11)	
	CDH23.3	8013	0	0	0	27	583	1823	2835	2332	406	6	0	0	(12)	
	CDH26.7	3709	0	0	0	2	179	823	1427	1171	108	0	0	0	(13)	
(14) Wind	WSAvg	4.3	4.5	4.6	5.1	5.0	4.5	4.2	3.8	3.6	4.0	4.0	3.9	4.4	(14)	
(15) Precipitation	PrecAvg	159	9	7	10	16	16	19	16	12	15	13	12	11	(15)	
	PrecMax	260	27	19	32	41	55	39	68	63	66	51	33	25	(16)	
	PrecMin	89	1	0	0	0	0	0	0	0	0	0	1	1	(17)	
	PrecStd	47	7	4	8	12	15	12	16	16	14	13	9	6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.2	7.9	16.2	26.3	32.7	37.1	37.7	38.0	32.7	24.2	16.0	8.9	(19)	
		MCWB	4.4	5.1	10.1	14.8	18.5	20.4	21.3	21.0	19.7	15.6	10.6	6.5	(20)	
	2%	DB	4.1	5.5	13.3	23.5	30.5	35.0	36.0	36.0	29.4	21.6	13.4	6.9	(21)	
		MCWB	2.8	3.2	8.1	14.0	18.3	20.4	21.3	20.6	18.3	13.8	10.1	5.0	(22)	
	5%	DB	2.7	3.9	11.0	21.0	28.7	33.1	34.6	34.4	27.4	19.5	11.3	5.2	(23)	
		MCWB	1.5	1.9	6.4	12.8	17.5	19.8	20.9	20.5	17.5	13.0	8.7	3.8	(24)	
	10%	DB	1.5	2.2	9.0	18.7	26.8	31.3	33.2	32.7	25.6	17.3	9.4	3.5	(25)	
		MCWB	0.4	0.7	5.4	11.5	16.8	19.1	20.6	20.2	16.8	12.0	7.3	2.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.7	5.8	10.6	15.7	21.0	23.6	25.2	24.1	21.4	16.8	12.1	7.2	(27)	
		MCDB	5.8	7.5	15.5	24.2	28.7	31.7	32.7	31.5	28.5	22.5	14.6	8.5	(28)	
	2%	WB	2.9	3.7	8.5	14.3	19.6	22.1	23.9	22.9	20.0	15.1	10.5	5.3	(29)	
		MCDB	4.0	5.2	12.7	22.1	27.4	30.9	31.3	31.5	26.5	19.6	12.8	6.6	(30)	
	5%	WB	1.6	2.0	6.9	13.3	18.7	21.1	22.8	22.0	18.8	13.6	8.9	3.8	(31)	
		MCDB	2.6	3.5	10.4	20.0	26.4	29.8	30.5	30.7	25.5	17.9	11.1	5.1	(32)	
	10%	WB	0.5	0.8	5.6	12.1	17.8	20.2	21.8	21.1	17.6	12.4	7.3	2.3	(33)	
		MCDB	1.4	2.1	8.6	17.7	25.0	28.6	30.1	29.9	24.2	16.6	9.3	3.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	7.3	8.2	10.8	10.5	11.1	11.9	13.2	12.2	10.3	7.4	5.2	(35)	
		MCDBR	5.9	7.7	11.4	14.0	12.5	14.0	14.0	15.0	14.2	13.2	9.9	5.7	(36)	
		MCWBR	5.1	5.8	7.1	6.3	4.8	5.0	5.1	5.3	6.1	6.9	6.8	4.9	(37)	
	5% WB	MCDBR	5.9	7.2	10.3	13.3	11.1	12.5	12.0	13.4	12.4	11.5	9.2	5.4	(38)	
		MCWBR	5.2	5.7	6.7	6.5	5.2	5.1	5.3	5.6	6.1	7.1	6.6	4.8	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.362	0.384	0.427	0.497	0.473	0.447	0.452	0.456	0.425	0.394	0.371	0.374	(40)	
	taud		2.113	2.066	2.049	1.921	2.062	2.196	2.194	2.156	2.182	2.217	2.190	2.097	(41)	
	Ebn at Noon		680	752	780	759	799	823	813	790	778	739	670	609	(42)	
	Edh at Noon		94	121	143	178	160	141	139	139	124	103	87	87	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.09	2.02	3.37	4.97	6.25	6.88	6.64	5.92	4.52	2.77	1.42	0.82	(44)	
	RadStd		0.17	0.24	0.35	0.41	0.50	0.49	0.32	0.29	0.30	0.27	0.21	0.14	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (2 neighbors)		+0.53	N/A	N/A	+1.46	N/A	N/A	N/A	N/A	+170	+115	

Nomenclature: See separate page